

Olist E-commerce Analysis

Business Insights & Recommendations Report

Chapter 1: Project Overview

1.1 Introduction

Olist E-commerce Analysis

The rapid growth of e-commerce has generated vast amounts of transactional data, making data analytics essential for improving business performance and customer experience. This project analyzes the Brazilian Olist E-commerce dataset to evaluate marketplace performance, customer behavior, product sales, and seller effectiveness.

Using SQL for data cleaning and business analysis and Tableau for interactive dashboard development, the project transforms raw transactional data into meaningful business insights that support data-driven decision-making.

1.2 Project Objective

The primary objective of this project is to analyze Olist's e-commerce operations by identifying key trends, evaluating marketplace performance, and providing actionable business recommendations.

The analysis focuses on answering the following business questions:

- How is the marketplace performing overall?
 - Which product categories generate the highest revenue?
 - How do customers behave across different regions?
 - Which sellers contribute the most to revenue?
 - How do freight costs and customer reviews influence business performance?
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1.3 Dataset Description

The analysis uses the **Brazilian Olist E-commerce Public Dataset**, which contains information about customers, orders, sellers, products, payments, reviews, and geographic locations.

The dataset represents real e-commerce transactions across Brazil and provides valuable insights into sales performance, logistics, customer satisfaction, and marketplace operations.

Main Tables Used

- Customers
- Orders
- Order Items

- Payments
- Reviews
- Products
- Sellers
- Product Category Translation
- Geolocation

1.4 Tools & Technologies

Tool	Purpose
SQL	Data cleaning, validation, business analysis
Tableau	Interactive dashboard development
Excel	Initial data inspection

1.5 Project Workflow

Raw Olist Dataset



Database Schema



Data Quality Assessment



Data Cleaning & Validation (SQL)



Exploratory Data Analysis (EDA)



Business KPI Analysis



Advanced Business Analysis



Star Schema Design



Interactive Dashboard Development (Tableau)



Business Insights & Recommendations

1.6 Dashboards Developed

This project consists of four interactive dashboards:

Executive Dashboard

Provides a high-level overview of revenue, orders, payment methods, regional performance, and key business KPIs.

Customer Insights Dashboard

Analyzes customer distribution, purchasing behavior, review scores, and customer satisfaction across different regions.

Product Performance Dashboard

Evaluates product category performance, pricing, freight costs, revenue contribution, and customer ratings.

Seller Performance Dashboard

Examines seller revenue, geographic distribution, freight costs, and seller contribution to marketplace performance

Chapter 2: Data Cleaning & Preparation

2.1 Data Preparation Overview

Before performing business analysis and developing dashboards, the raw Olist dataset was thoroughly cleaned and validated to ensure data accuracy, consistency, and reliability. Since the dataset consisted of multiple related tables, several data quality checks and transformations were performed using SQL.

The objective of the data preparation process was to eliminate data quality issues, validate business rules, and prepare a clean analytical dataset for visualization in Tableau.

2.2 Data Cleaning Process

The following data preparation activities were completed:

Missing Value Analysis

Each table was examined for missing (NULL) values to identify incomplete records.

The following fields were carefully validated:

- Product Category
- Product Description

- Product Dimensions
- Review Messages
- Order Delivery Dates
- Payment Information
- Customer Information
- Seller Information

Appropriate validation was performed to determine whether missing values represented genuine business scenarios or data quality issues.

✅ Duplicate Record Validation

Duplicate records were identified using SQL queries to ensure the uniqueness of primary keys and transactional records.

The following tables were checked:

- Customers
- Orders
- Order Items
- Payments
- Reviews
- Products
- Sellers

Duplicate records were removed wherever necessary to maintain data integrity.

✅ Data Type Validation

Several columns were validated to ensure appropriate data types for analysis.

Examples include:

- Order Date → Date
- Delivery Date → Date
- Price → Decimal
- Freight Value → Decimal
- Review Score → Integer

This ensured accurate calculations and aggregations during dashboard development.

✔ Business Rule Validation

Business validation rules were applied to verify the consistency of transactional data.

Examples include:

- Orders with missing delivery dates
- Orders marked as delivered without shipment dates
- Zero payment values
- Invalid product dimensions
- Missing product categories
- Invalid freight values

These checks helped identify anomalies and improve data quality.

2.3 Data Integration

The project dataset consisted of multiple relational tables linked through primary and foreign keys.

The main relationships included:

Table	Join Key
Customers ↔ Orders	Customer ID
Orders ↔ Order Items	Order ID
Orders ↔ Payments	Order ID
Orders ↔ Reviews	Order ID
Order Items ↔ Products	Product ID
Order Items ↔ Sellers	Seller ID

These relationships enabled a comprehensive analysis across customers, products, sellers, payments, and orders.

2.4 Data Quality Checks

The following SQL-based quality checks were performed before analysis:

- Record count validation
- Duplicate detection
- Missing value analysis

- Blank value validation
- Business rule validation
- Payment consistency checks
- Product information validation
- Review data verification
- Seller information validation

These checks ensured the analytical dataset was accurate and suitable for reporting.

2.5 Data Preparation Outcome

After completing the data cleaning and validation process:

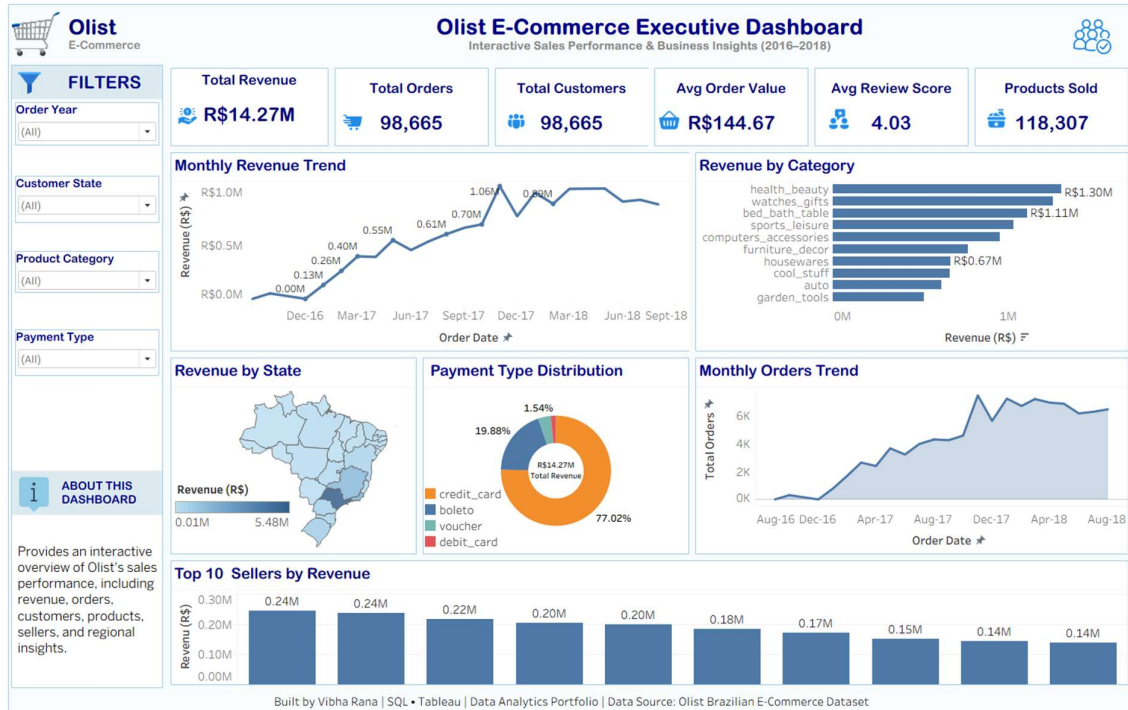
- The dataset was successfully cleaned and standardized.
- Invalid and duplicate records were addressed.
- Business rules were validated.
- Data relationships were verified.
- The cleaned dataset was prepared for interactive dashboard development in Tableau.

This structured preparation process improved the reliability and accuracy of the business insights presented throughout the project.

Chapter 3: Executive Dashboard Analysis

3.1 Dashboard Overview

The Executive Dashboard provides a comprehensive overview of Olist's marketplace performance by presenting key business metrics, sales trends, product performance, customer activity, seller contribution, payment methods, and regional revenue distribution. It enables decision-makers to monitor business performance and identify opportunities for growth through an interactive interface with dynamic filters.



3.2 Dashboard Objective

The primary objective of this dashboard is to monitor the overall performance of the Olist marketplace by tracking revenue, customer orders, products sold, payment preferences, top-performing product categories, seller performance, and regional sales trends.

3.3 Key Performance Indicators

KPI	Value
Total Revenue	R\$14.27M
Total Orders	98,665
Total Customers	98,665

KPI	Value
Average Order Value	R\$144.67
Average Review Score	4.03 / 5
Products Sold	118,307

3.4 Key Business Insights

1. Strong Revenue Performance

The marketplace generated a total revenue of **R\$14.27 million**, reflecting strong overall business performance and sustained customer demand throughout the analysis period.

2. High Marketplace Activity

The platform processed **98,665 customer orders** with **118,307 products sold**, indicating that many orders included multiple products and demonstrating healthy transaction activity.

3. Positive Customer Satisfaction

The average customer review score of **4.03 out of 5** suggests that customers were generally satisfied with their shopping experience, products, and delivery services.

4. Monthly Revenue Shows Consistent Growth

The Monthly Revenue Trend shows a steady increase from late 2016 through 2017, reaching peaks of approximately **R\$1.0 million** per month before stabilizing. This indicates sustained business growth over time.

5. Order Volume Increased Alongside Revenue

The Monthly Orders Trend closely follows the revenue trend, showing a significant increase in order volume during 2017 and early 2018. This suggests that revenue growth was supported by increased customer purchasing activity rather than isolated high-value transactions.

6. Health & Beauty Leads Product Revenue

Among all product categories, **Health & Beauty** generated the highest revenue, followed by **Watches & Gifts** and **Bed & Bath Table**. These categories represent the marketplace's strongest revenue contributors.

7. Revenue Is Concentrated in Specific States

The Revenue by State map shows that revenue is unevenly distributed across Brazil, with a few states contributing a significantly larger share of total sales than others. This highlights key regional markets for the business.

8. Credit Card Is the Preferred Payment Method

The Payment Type Distribution indicates that **Credit Card accounts for approximately 77%** of all payments, making it the dominant payment method. Boleto contributes nearly **20%**, while Voucher and Debit Card represent only a small share.

9. Revenue Is Distributed Across Multiple Sellers

The Top 10 Sellers by Revenue chart shows that the highest-performing sellers generated revenues ranging from approximately **R\$0.14M to R\$0.24M**. While several sellers contribute strongly, no single seller overwhelmingly dominates the marketplace, suggesting a relatively balanced seller ecosystem.

3.5 Business Recommendations

Based on the dashboard analysis, the following recommendations are proposed:

- Increase marketing efforts for the **Health & Beauty** and **Watches & Gifts** categories to maximize revenue from high-performing products.
 - Focus regional marketing campaigns on states with strong sales while identifying opportunities to improve performance in lower-revenue regions.
 - Continue optimizing the **credit card payment experience**, as it is the most widely used payment method.
 - Maintain high customer satisfaction by monitoring product quality and delivery performance to sustain review scores above 4.0.
 - Use monthly sales and order trends to forecast inventory requirements and plan promotional campaigns during high-demand periods.
 - Regularly monitor top sellers and provide support programs to help emerging sellers improve their marketplace performance.
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3.6 Executive Summary

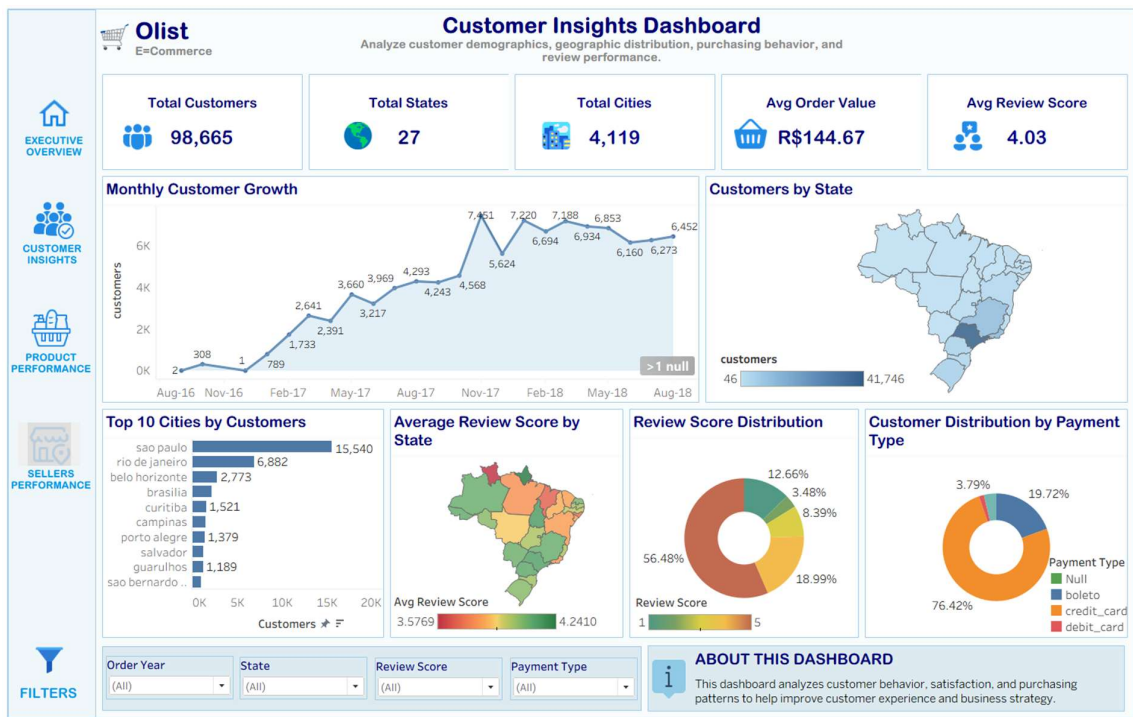
The Executive Dashboard highlights strong marketplace performance with **R\$14.27 million in revenue**, nearly **99 thousand customer orders**, and **118 thousand products sold**. Revenue and order volumes demonstrate steady growth over the analysis period, while customer satisfaction remains positive with an average review score of **4.03**. Product category analysis identifies **Health & Beauty** as the leading revenue generator, and payment analysis confirms

Credit Card as the preferred payment method. Overall, the dashboard provides a comprehensive overview of business performance and supports informed strategic decision-making through interactive visualizations and key performance metrics.

Chapter 4: Customer Insights Dashboard Analysis

4.1 Dashboard Overview

The Customer Insights Dashboard provides a comprehensive analysis of customer demographics, purchasing behavior, geographic distribution, payment preferences, and customer satisfaction. It enables businesses to understand where customers are located, how they purchase, and how satisfied they are with their shopping experience. These insights help support customer-centric strategies and improve overall business performance.



4.2 Dashboard Objective

The primary objective of this dashboard is to analyze customer behavior and satisfaction by monitoring customer growth, geographic distribution, review scores, payment preferences, and purchasing trends. It provides valuable insights that support customer retention, market expansion, and improved customer experience.

4.3 Key Performance Indicators (KPIs)

KPI	Value
Total Customers	98,665
Total States	27
Total Cities	4,119
Average Order Value	R\$144.67
Average Review Score	4.03 / 5

4.4 Key Business Insights

Insight 1: Strong Customer Base Across Brazil

The marketplace serves **98,665 customers** across **27 Brazilian states** and **4,119 cities**, demonstrating a broad customer reach and nationwide presence.

Insight 2: Customer Growth Increased Over Time

The Monthly Customer Growth chart shows a steady increase in new customers from 2016 through 2018. Customer acquisition accelerated significantly during 2017 before stabilizing at consistently high monthly levels.

Insight 3: São Paulo Has the Largest Customer Base

Among all cities, **São Paulo** has the highest number of customers (**15,540**), followed by **Rio de Janeiro (6,882)** and **Belo Horizonte (2,773)**. These cities represent the marketplace's largest urban customer markets.

Insight 4: Customer Distribution Is Concentrated in Key States

The Customers by State map indicates that customer activity is concentrated in a few highly populated states, while several regions have comparatively lower customer counts. This suggests opportunities for targeted regional expansion.

Insight 5: Customer Satisfaction Is Generally Positive

The marketplace maintains an **average review score of 4.03**, indicating that customers are generally satisfied with their purchasing experience.

Insight 6: Five-Star Reviews Dominate Customer Feedback

The Review Score Distribution shows that **5-star ratings account for approximately 56.48%** of all reviews, while lower ratings represent a much smaller share. This reflects a high level of customer satisfaction and positive purchasing experiences.

Insight 7: Credit Card Is the Preferred Payment Method

The Customer Distribution by Payment Type chart shows that **76.42%** of customers use **Credit Card** payments, followed by **Boleto (19.72%)**, while **Debit Card (3.79%)** accounts for only a small percentage of transactions.

Insight 8: Consistent Average Order Value

The **Average Order Value of R\$144.67** indicates relatively stable customer spending across the marketplace and provides a useful benchmark for evaluating future sales performance.

4.5 Business Recommendations

Based on the customer analysis, the following recommendations are proposed:

- Expand marketing campaigns in high-growth cities while increasing customer acquisition efforts in underrepresented regions.
 - Develop customer loyalty programs to encourage repeat purchases and improve customer retention.
 - Continue focusing on customer service and delivery quality to maintain high review scores.
 - Enhance the checkout experience by optimizing credit card payment processes while promoting alternative payment options.
 - Use customer geographic insights to support region-specific promotions and localized marketing campaigns.
 - Monitor customer growth trends regularly to forecast demand and allocate business resources effectively.
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4.6 Customer Dashboard Summary

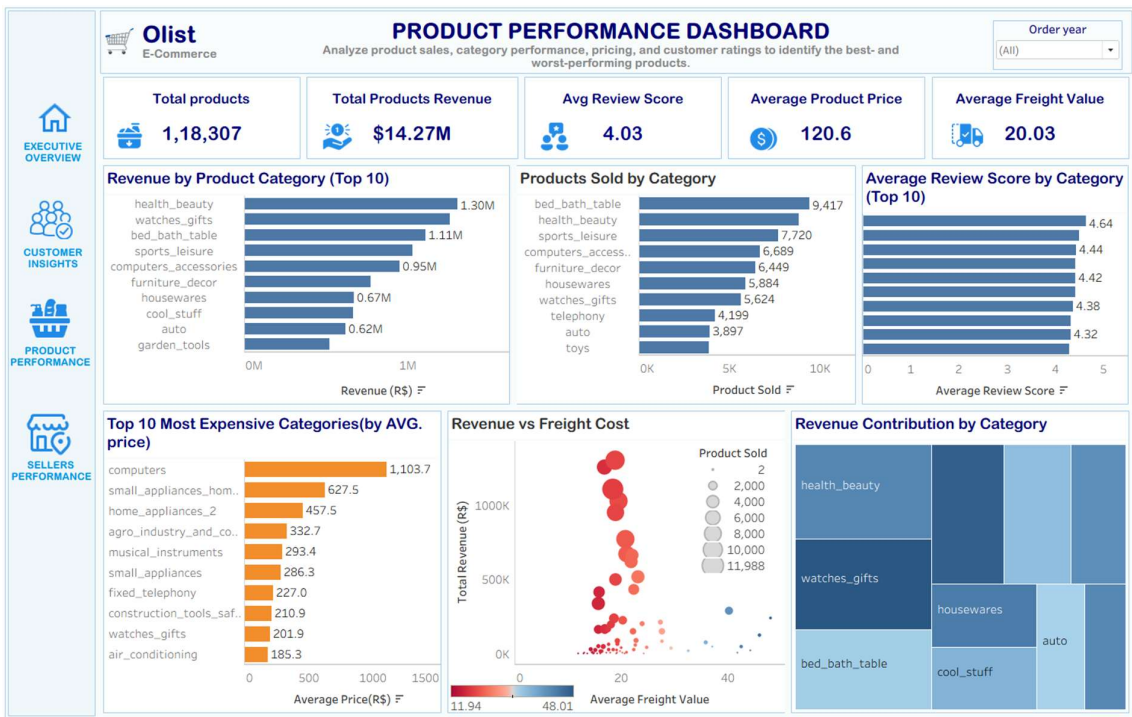
The Customer Insights Dashboard demonstrates that Olist has built a strong nationwide customer base with **98,665 customers** across **27 states** and **4,119 cities**. Customer acquisition has grown steadily over time, with major metropolitan areas such as São Paulo driving a significant share of marketplace activity. High customer satisfaction, reflected by an average review score of **4.03** and a majority of **5-star reviews**, indicates a positive shopping experience. Additionally, the dominance of credit card payments highlights customer

preferences during checkout. These insights provide valuable guidance for improving customer engagement, expanding into new markets, and supporting long-term business growth.

Chapter 5: Product Performance Dashboard Analysis

5.1 Dashboard Overview

The Product Performance Dashboard provides a detailed analysis of product sales, revenue, pricing, freight costs, customer ratings, and category performance. It helps identify high-performing product categories, pricing trends, and the relationship between freight costs and revenue, enabling better product and inventory management decisions.



5.2 Dashboard Objective

The primary objective of this dashboard is to evaluate product performance by analyzing revenue, sales volume, pricing, freight charges, and customer ratings across different product categories. The insights help identify top-performing products and support data-driven merchandising strategies.

5.3 Key Performance Indicators (KPIs)

KPI	Value
Total Products Sold	118,307
Total Product Revenue	R\$14.27M
Average Review Score	4.03 / 5
Average Product Price	R\$120.60
Average Freight Value	R\$20.03

5.4 Key Business Insights

Insight 1: Strong Product Sales Performance

A total of **118,307 products** were sold, generating **R\$14.27 million** in revenue. This highlights strong customer demand across multiple product categories.

Insight 2: Health & Beauty Is the Highest Revenue-Generating Category

The **Health & Beauty** category generated the highest revenue (**approximately R\$1.30M**), followed by **Watches & Gifts** and **Bed & Bath Table**. These categories are the marketplace's strongest revenue contributors.

Insight 3: Bed Bath Table Leads in Product Sales

The **Bed Bath Table** category recorded the highest number of products sold (**9,417 units**), followed by **Health & Beauty (8,944)** and **Sports & Leisure (7,720)**. High sales volume indicates strong customer demand for these categories.

Insight 4: Premium Categories Have Higher Average Prices

The **Computers** category has the highest average product price (**approximately R\$1,103.70**), making it the most premium category in the marketplace. Other categories such as **Small Home Appliances** and **Home Appliances** also have significantly higher average prices than the marketplace average.

Insight 5: Customer Satisfaction Remains High Across Top Categories

Several product categories maintain average review scores above **4.3**, with the highest-rated category achieving an average score of **4.64**. This indicates consistently positive customer experiences across top-performing categories.

Insight 6: Average Product Pricing Is Balanced

The overall **average product price of R\$120.60** suggests that the marketplace primarily serves customers purchasing products within an affordable price range while still offering premium products in selected categories.

Insight 7: Freight Costs Show a Positive Relationship with Revenue

The Revenue vs Freight Cost scatter plot indicates that higher-revenue categories often have moderately higher freight costs. However, revenue growth is not driven solely by freight charges, suggesting that product demand plays a more significant role.

Insight 8: Revenue Contribution Is Concentrated Among Key Categories

The treemap shows that categories such as **Health & Beauty**, **Watches & Gifts**, **Bed Bath Table**, **Housewares**, and **Cool Stuff** contribute a substantial share of total marketplace revenue, while many other categories contribute smaller portions.

5.5 Business Recommendations

Based on the product analysis, the following recommendations are proposed:

- Increase inventory levels for high-demand categories such as **Health & Beauty**, **Bed Bath Table**, and **Sports & Leisure** to reduce stock-out risks.
 - Continue promoting high-revenue categories through targeted marketing campaigns to maximize sales.
 - Monitor premium-priced categories to ensure pricing remains competitive while maintaining profitability.
 - Optimize freight costs by improving logistics and negotiating shipping rates, especially for high-volume categories.
 - Leverage highly rated product categories in promotional campaigns to strengthen customer trust and increase conversions.
 - Regularly review category performance to identify emerging trends and adjust inventory planning accordingly.
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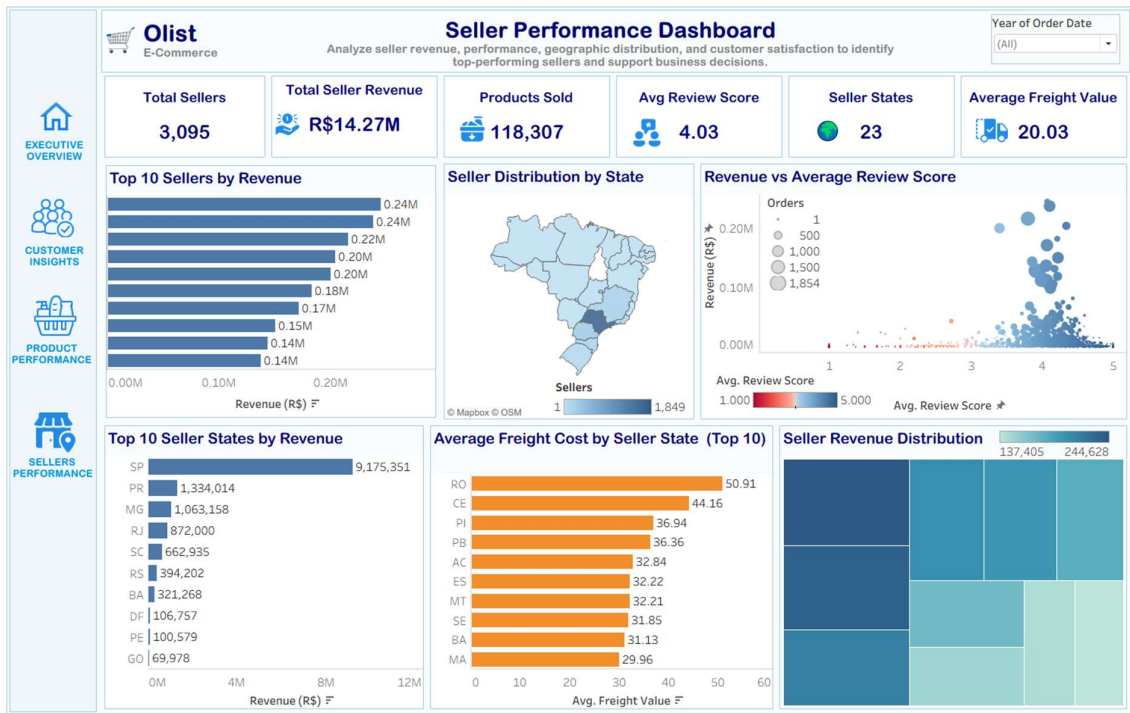
5.6 Product Dashboard Summary

The Product Performance Dashboard highlights the marketplace's strong product portfolio, with **118,307 products sold** generating **R\$14.27 million** in revenue. Categories such as **Health & Beauty**, **Watches & Gifts**, and **Bed Bath Table** are key drivers of business performance. The dashboard also demonstrates positive customer satisfaction, balanced product pricing, and valuable insights into freight costs and category contribution. These findings support strategic decisions related to inventory management, pricing, logistics, and product marketing, ultimately helping improve overall marketplace performance.

Chapter 6: Seller Performance Dashboard Analysis

6.1 Dashboard Overview

The Seller Performance Dashboard provides a comprehensive analysis of seller performance across the Olist marketplace. It evaluates seller revenue, geographic distribution, freight costs, customer ratings, and product sales to identify top-performing sellers and regional trends. These insights support business decisions related to seller management, logistics, and marketplace growth.



6.2 Dashboard Objective

The primary objective of this dashboard is to evaluate seller performance by analyzing revenue generation, seller distribution, customer satisfaction, freight costs, and regional contributions.

It helps identify high-performing sellers and supports strategies to improve marketplace efficiency.

6.3 Key Performance Indicators (KPIs)

KPI	Value
Total Sellers	3,095
Total Seller Revenue	R\$14.27M
Products Sold	118,307
Average Review Score	4.03 / 5
Seller States	23
Average Freight Value	R\$20.03

6.4 Key Business Insights

Insight 1: Strong Seller Network

The marketplace operates with **3,095 active sellers** across **23 Brazilian states**, creating a diverse and competitive seller ecosystem.

Insight 2: Sellers Generated Significant Revenue

The seller network collectively generated **R\$14.27 million** in revenue through the sale of **118,307 products**, demonstrating the marketplace's strong commercial performance.

Insight 3: São Paulo Is the Leading Seller State

The **Top Seller States by Revenue** chart shows that **São Paulo (SP)** generated the highest seller revenue (**approximately R\$9.18 million**), significantly outperforming all other states. Paraná (PR) and Minas Gerais (MG) are the next strongest contributors.

Insight 4: Revenue Is Concentrated Among Top Sellers

The **Top 10 Sellers by Revenue** chart shows that the highest-performing sellers generated between **R\$0.14M and R\$0.24M** in revenue. While several sellers contribute strongly, revenue

is distributed across multiple high-performing sellers rather than being dominated by a single seller.

Insight 5: High Review Scores Are Associated with Better Revenue

The **Revenue vs Average Review Score** scatter plot indicates that sellers with review scores close to **4 and above** generally achieve higher revenue. This suggests that maintaining good customer satisfaction contributes to stronger seller performance.

Insight 6: Freight Costs Differ Across Seller States

The **Average Freight Cost by Seller State** chart reveals noticeable regional differences in shipping costs. **Rondônia (RO)** has the highest average freight value (**R\$50.91**), while other states maintain comparatively lower shipping costs.

Insight 7: Seller Presence Is Concentrated in Specific Regions

The **Seller Distribution by State** map shows that seller activity is concentrated in a limited number of states, indicating that marketplace operations are stronger in certain regions than others.

Insight 8: Revenue Contribution Is Distributed Across Multiple Sellers

The **Seller Revenue Distribution** treemap demonstrates that revenue is shared among several major sellers, reducing dependence on a single seller and contributing to a more balanced marketplace.

6.5 Business Recommendations

Based on the seller analysis, the following recommendations are proposed:

- Strengthen seller acquisition programs in underrepresented states to expand marketplace coverage.
- Provide training and performance support for new and medium-performing sellers to improve sales performance.
- Encourage sellers to maintain high customer service standards, as better review scores are associated with higher revenue.
- Optimize logistics and shipping operations in states with higher freight costs to improve profitability.
- Recognize and reward top-performing sellers through incentive programs to encourage long-term engagement.

- Continuously monitor seller performance using KPIs and dashboards to identify improvement opportunities and maintain marketplace competitiveness.
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6.6 Seller Dashboard Summary

The Seller Performance Dashboard highlights a strong marketplace supported by **3,095 active sellers** operating across **23 states**. Sellers collectively generated **R\$14.27 million** in revenue while selling over **118,000 products**. The analysis shows that **São Paulo** is the leading contributor to seller revenue, customer satisfaction remains high with an average review score of **4.03**, and freight costs vary across regions. These insights provide valuable guidance for improving seller performance, optimizing logistics, and expanding marketplace operations.

Chapter 7: Overall Business Recommendations

Based on the analysis of sales performance, customer behavior, product performance, and seller insights, the following business recommendations are proposed to improve marketplace growth, customer satisfaction, and operational efficiency.

7.1 Expand High-Performing Product Categories

Product categories such as **Health & Beauty**, **Watches & Gifts**, and **Bed & Bath Table** consistently generated high revenue and sales volumes. Olist should prioritize these categories by expanding inventory, introducing new product variations, and increasing promotional activities to maximize revenue.

7.2 Increase Customer Acquisition in Emerging Regions

Customer and revenue distribution indicate that business performance is concentrated in a few Brazilian states and cities. Expanding targeted marketing campaigns in underrepresented regions can help attract new customers and increase market penetration.

7.3 Strengthen Customer Loyalty

Although customer satisfaction is high, maintaining long-term customer relationships is essential. Implementing loyalty programs, personalized offers, and reward systems can encourage repeat purchases and improve customer retention.

7.4 Support Seller Development

The marketplace includes more than **3,000 sellers**, but revenue generation varies significantly among them. Providing seller training, performance insights, and incentive programs can help smaller sellers improve their sales and contribute more effectively to marketplace growth.

7.5 Improve Logistics and Freight Efficiency

Freight costs vary across seller states, indicating opportunities to optimize shipping operations. Collaborating with logistics partners, improving warehouse distribution, and optimizing delivery routes can reduce shipping costs and improve customer satisfaction.

7.6 Maintain High Customer Satisfaction

An average review score above **4.0** demonstrates positive customer experiences. Olist should continue monitoring product quality, delivery performance, and customer support to maintain high satisfaction levels and strengthen brand reputation.

7.7 Leverage Data-Driven Decision Making

Interactive dashboards should be regularly used to monitor key business metrics such as revenue, customer growth, product performance, and seller performance. Continuous monitoring enables faster decision-making and helps identify new business opportunities.

Chapter 8: Conclusion

8.1 Project Conclusion

This project analyzed the **Brazilian Olist E-commerce Dataset** using **SQL** for data cleaning and analysis and **Tableau** for interactive dashboard development. The objective was to transform raw transactional data into meaningful business insights that support strategic decision-making.

The analysis covered four key business areas:

- **Executive Performance**
- **Customer Insights**
- **Product Performance**
- **Seller Performance**

The dashboards revealed strong marketplace performance with **R\$14.27 million in revenue**, **98,665 customers**, **118,307 products sold**, and **3,095 active sellers**. Customer satisfaction remained high with an average review score of **4.03**, while analysis identified top-performing product categories, leading seller regions, and customer purchasing trends.

Overall, the project demonstrates how business intelligence tools can convert large volumes of data into actionable insights that improve operational efficiency, enhance customer experience, and support long-term business growth.

Business Impact

The insights generated from this project can help business stakeholders to:

- Improve strategic decision-making through interactive dashboards.
- Identify high-performing products, sellers, and customer segments.
- Optimize inventory and logistics planning.
- Enhance customer satisfaction and retention.
- Support future business expansion through data-driven strategies.

Skills Demonstrated

This project demonstrates practical skills in:

- SQL Data Cleaning & Validation
- Exploratory Data Analysis (EDA)
- Relational Database Querying
- Business Intelligence
- Tableau Dashboard Development
- KPI Design & Performance Monitoring
- Data Visualization
- Business Insight Generation
- Business Recommendation Development
- Analytical Problem Solving

References

- **Dataset:** Brazilian Olist E-commerce Public Dataset (Kaggle)
 - **Visualization Tool:** Tableau Public/Desktop
 - **Database Tool:** Microsoft SQL Server
 - **Documentation:** Tableau and SQL official documentation
-